

## **Xylem & plumbing**

The hierarchical arrangement of the membranes in the xylem tissue effectively amplifies the available membrane area for filtration, providing high flow rates. Xylem filters were prepared by simply removing the bark of pine tree branches and inserting the xylem tissue into a tube.

### ***How the Xylem Tissue in Plants Inspired a Cost Effective Filtration System***

Water is necessary for all life on our planet. It is the catalyst and solvent for most of life's chemistry. Since water goes so well with everything, it is also common for many microbes, dirt and other particles to dissolve in it and contaminate it for human consumption. **Plants use the xylem transportation system for cycling water from the ground through its body for use in hydration and photosynthesis. This tissue has an inbuilt filtration system which filters impurities and small to moderate sized particles from the water.** MIT is currently developing a low cost modular filtration system that can be used as an end point filtration system for resource poor and developing countries. Small pieces of sapwood from trees can be used for the same which can help in wide scale adoption of the technology.

Inspired by: Xylem tissue in plants

📍 **Where:** Massachusetts, USA

🔄 **Innovation / Function:** Cost effective filtration system

✍️ **Who:** MIT



Xylem's manufacturing facility in Emmaboda, Sweden, has a new water treatment plant that recycles the plant's process water. The plant helps ensure a continuous supply of safe water, even in times of water scarcity, using Xylem's water reuse technology.

Xylem's Emmaboda factory is the largest wastewater pump factory in the world, where 1,200 employees produce around 170,000 pumps each year. The factory's on-site water treatment plant, inaugurated in 2022, can recycle 20 million liters of water each year. The recycled water is used in pump test tanks, for cooling in the foundry, and to dilute cutting fluids in machining processes.

The initiative follows years of alarmingly low water levels in southern Sweden, which are expected to become even worse due to climate change. Through its new treatment plant, Xylem will ensure water remains available regardless of external factors.

“Following the drought in 2018, our Emmaboda factory was three days away from having to stop production as the city, understandably, needed to prioritize water for the more immediate needs of its population,” says Roger Sandström, Senior Director, Pumps & Circulators, Europe. “Going forward, it is likely that the region will keep facing lower rainwater levels due to climate change. As such, we are excited to launch the Emmaboda treatment plant, which will greatly reduce our usage of regional water supplies.”

### **A unique water reuse solution combining Xylem technologies**

Designed and supplied by Xylem, the treatment plant collects, treats, purifies, and reuses process and stormwater. The plant demonstrates how companies can achieve superior water quality and consistent water flow with Xylem’s water reuse technologies, at low cost and without extensive training. The solution works by leveraging natural surroundings and combining more than 20 Xylem technologies into a compact system.

When used process water leaves the factory, it first passes through a sludge tank to remove heavy particles. It is then collected, along with stormwater, in nearby ponds on the property. As the water sits in the ponds, it undergoes natural settling, as well as biological purification from cultivated herbs. From the ponds, water is pumped to the treatment plant using intelligent, energy-efficient Flygt Concertor pumps.

At the plant, the water passes through five treatment stages – **oxidation, flocculation, clarification, sand filtration, and UV disinfection**. The incoming water to the plant is analyzed to set appropriate chemical dosing in the first two stages. In the clarification stage, the Leopold Texler lamella clarifier is used to provide a large settling area within a small footprint, removing 98% of impurities. The clarified water then passes through a sand bed to filter out fine particles and contaminants and remove color. Finally, a Wedeco Spektron UV disinfection system provides a microbacterial barrier that eliminates more than 99.99% of all pathogens in just a fraction of a second.

Once treated, the water is stored in a clean water reservoir, ready for Lowara variable speed booster sets to deliver it to the factory based on water demand. The treatment process runs automatically while WTW water analyzers, intelligent equipment controls, and a Xylem SCADA system work together to enable real-time, remote monitoring and alerts.